
Pattern Arithmetic: Verdicts from Bowls and Paths

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Abstract

A companion paper, *Pattern Arithmetic: Typed Algebra of Bounded Units on a Phase-Colored Sphere*, defines a unit of truth $x = (r, \hat{n}, \chi, \tau)$ on a sphere $\mathcal{S}$ and an algebra over such units — gather ($\oplus$), remove ($\ominus$), mix ($\boxplus$), walk (T), and offset — together with an admissibility test that decides which sorts τ may use which operators truthfully. That paper builds the algebra and stops at the reading: a bowl of units may be mixed or walked, and the result is a state. It does not say how a state, or a bowl, becomes a *decision*.

This paper adds the missing step. A shaman, a tribal meteorologist, or a market analyst holds a handful of signs and must arrive at one of a small number of conclusions — water or no water, storm or calm, rise or fall. We show that the algebra already contains two distinct mechanisms for this, corresponding to the two readings the first paper leaves undemonstrated: an order-free reading, in which evidence accumulates as phasors and a balanced case returns no answer rather than a forced one; and an order-dependent reading, in which evidence is walked as a sequence of actors to a landing compared against seated names. We give a rule — the *swap test* — for which reading a given body of evidence calls for, and identify four components a decision needs beyond what the algebra supplies: a comparison rule, a selection rule for which signs enter and in what order, a principled abstention, and a place for a lexicon of conclusions to be learned rather than declared. This paper is a framework: it specifies these four components precisely enough to be filled in, and works one example — the diviner’s shells — by hand. It does not report an evaluation against data, and says exactly where that evaluation would have to be made.

1 Introduction

A shaman throws ten shells and reads a future from them. He may throw twenty and read it better, or forty better still — but unless he says so, no one else knows what he saw. The shells are evidence; the saying is the verdict. Pattern Arithmetic gives a precise account of the shells. It has none, so far, of the saying.

This is deliberate on the first paper’s part, not an oversight. That paper’s discipline was to make the algebra trustworthy before asking it to decide anything: an admissibility test that refuses sorts with no honest map onto the operators, a type table that says which combinations are legal, and a running insistence that a walk’s landing is compared to a name, never confused with one. Asking that paper to also commit to a decision rule would have asked it to defend a claim it was not ready to make.

It is ready now. This paper takes the atom, the bowl, and the path as given — cited, not re-derived — and asks the next question: given a bowl of evidence, what is the answer?

1.1 A human can hold a handful; a machine need not stop there

A diviner’s working memory bounds how many shells, how many birds, how many signs can be weighed against one question at once. Nothing in the algebra of the first paper imposes that bound. The claim of this paper is narrower than it might sound: not that more evidence is always better, but that *the mechanism for combining evidence does not change shape as the amount of evidence grows* — the same gather, the same two readings, apply to ten shells or ten thousand sensor readings. Where that mechanism strains under scale (§4.3) is stated plainly rather than smoothed over.

2 What Is Inherited

We use, without re-deriving, the following from the companion paper. Equation and axiom numbers refer to that paper.

- The atom $x = (r, \hat{n}, \chi, \tau) \in \mathcal{S}$, with $r \leq 1$ a normalized strength, $\hat{n} \in S^2$ a direction, $\chi \in [0, 2\pi)$ a phase, and τ a sort tag that gates which operators may be truthfully applied (Axiom 1; Axiom 0).
- Gather, $\oplus$: union on finite subsets of $\mathcal{S}$, sort-blind, idempotent, lossless, undone by $\ominus$ (Axiom 4).
- Mix, $\boxplus$: a phasor reading of a bowl whose members share a direction and a sort, $\boxplus(B) = (\min(1, |z_B|), \hat{n}, \arg z_B, \tau)$ with $z_B = \sum_{x \in B} r_x e^{i\chi_x}$; commutative, non-associative, lossy, uninvertible (Axiom 5).
- The walk, T : an ordered composition $T_{w_n} \circ \cdots \circ T_{w_1}(x)$ of actor-parameterized rotations and phase shifts; non-commutative in general, and many-to-one, so a landing alone does not determine the walk that reached it (Axiom 6; §3).
- *Levels are closed*: a unit carries no representation of what it constitutes, and a bowl carries no representation of the reading that will be applied to it. Whatever a lower unit would need to know about a higher one — which units were taken as

co-present, over what interval, under what question — is a constituting fact of the higher level, not a coordinate of the atom (companion paper, §2.2).

Two consequences of level closure matter enough here to restate. First, a bowl that saturates under Mix is often a level that has been flattened — repeated samples of one unit, mistaken for many units at one level (companion paper, §2.7). Second, the atom stays a four-tuple; grouping criteria such as a time window or a spatial radius live outside it, deciding bowl membership before the algebra runs rather than riding inside any single unit.

2.1 A substitution: bounding aggregated strength

The companion paper’s Axiom 1 names its clip, $r' = \min(1, |z|)$, as a choice, and names an order-preserving bijection as the alternative a setting that aggregates at scale should weigh — naming this paper’s own use as exactly such a setting, rather than leaving that identification for the reader to make. This paper takes that invitation: wherever a strength is reported from a phasor sum for reading purposes (Reading A’s own verdict, and a reference atom’s own reported strength), it substitutes

$$r' = \frac{|z|}{K + |z|}, \quad (1)$$

the K -calibrated case of the general Hill form $x^n/(K^n + x^n)$ at $n = 1$, for the companion paper’s clip. K is chosen per domain, not fixed here: it sets where the half-confidence point sits on the scale of that domain’s own evidence. Held at $n = 1$, this does not trade away the reason a bijection was wanted in the first place — the tail behaviour $1 - r' \approx K/|z|$ for large $|z|$ still lets arbitrarily large aggregates be distinguished, only at a correspondingly large scale set by K . Raising n above 1 is not free in the same way: it steepens the transition near K at the cost of exactly that tail, reintroducing practical indistinguishability at a smaller scale rather than removing it, and is not adopted here.

This substitution leaves $\bar{R}_c$ (Equation (4)) untouched: that quantity already normalizes by the raw, uncapped $|z|$, and was built for the same reason this substitution is — to stop a bound from erasing a real difference in agreement.

3 Two Readings of One Bowl

Given a bowl $B = \{x_1, \dots, x_n\}$ of evidence bearing on one question, the companion paper currently defines two readings that turn B into a single state: phasor Mix and ordered transformation. This is not a claim that no other reading could be defined; it is the pair that paper supplies today, and the pair this paper uses. We name them here because the first paper defines both and demonstrates only one.

3.1 Reading B: the walk

Seat the question as a state q . Apply the members of B as actors, in some order π :

$$\text{land}_\pi(q) = T_{x_{\pi(n)}} \circ \dots \circ T_{x_{\pi(1)}}(q).$$

The verdict is the landing, compared against seated names (the storm’s calm and storm pins are an instance of this). Order matters by construction: $T_a T_b \neq T_b T_a$ in general, so

$\text{land}_\pi \neq \text{land}_{\pi'}$ for $\pi \neq \pi'$ is the expected case, not a failure. This reading is fully worked in the companion paper’s storm example, which we do not repeat.

3.2 Reading A: the phasor sum

Rotate every member of B onto the question’s axis $\hat{n}_q$ (a rotation, which preserves r and χ), then read the whole bowl at once:

$$\boxplus(B) = \left(\min(1, |z_B|), \hat{n}_q, \arg z_B, \tau \right), \quad z_B = \sum_{x \in B} r_x e^{i\chi_x}.$$

Order plays no role: z_B is a sum, and sums do not care about the order of their terms. The verdict’s *confidence* is $|z_B|$ ’s image under the cap, and its *direction* is $\arg z_B$, read against a lexicon of conclusions seated at particular phases (e.g. $\chi = 0$ for “water,” $\chi = \pi$ for “no water”). Agreeing evidence lengthens the sum; opposing evidence shortens it; evenly opposed evidence returns $z_B = 0$ — black, and a principled abstention rather than a forced choice (companion paper, Axiom 2).

3.3 The swap test

Which reading applies to a given B is decided by one question: does exchanging the order in which two members of B arrived change the answer?

- If **no** — the shells thrown together, ten readings taken at one instant — B has no order to respect, and Reading A is the correct mechanism. Applying Reading B here manufactures an order-dependence that is not in the evidence.
- If **yes** — birds fleeing, then a wind shift, then a temperature drop, where the same three signs in a different sequence mean something else — B ’s order is itself information, and Reading A would discard it. Reading B is required.

The two readings are not competing hypotheses about one mechanism; they are the correct tool for two different shapes of evidence, and the swap test is how a system — human or otherwise — tells which shape it is holding.

3.4 The readings compose

Nothing prevents using both on one episode. A morning’s observations may arrive in a meaningful sequence (Reading B over the morning) and also include several simultaneous signs at one moment within it (Reading A over that moment’s bowl, itself one actor in the larger walk). This is level closure in practice (§2): a Reading A verdict is a state, and a state may act as w in a T -path, exactly as any other seated unit may.

4 Four Components the Algebra Does Not Supply

Both readings above take a bowl and return a state. Neither, on its own, returns a decision a person or a system can act on. Four components stand between a state and a verdict, and none of them is supplied by $\oplus$, $\boxplus$, T , or offset.

4.1 A comparison rule

The companion paper’s storm section says a ball falls *near* the calm pin or the storm pin, and does not define near. A working comparison rule must state, at minimum: which coordinates of the landing are read, what metric is used on those coordinates, and whether τ must match exactly or may differ.

Two states are compared only if $\tau_A = \tau_B$; τ gates comparison the same way it gates every other operator. Given that, a distance is taken over all three geometric coordinates, each measured the way the companion paper already measures it elsewhere rather than by naive coordinate subtraction:

$$d(A, B)^2 = w_r (r_A - r_B)^2 + w_{\hat{n}} \arccos(\hat{n}_A \cdot \hat{n}_B)^2 + w_{\chi} \left(\min(|\chi_A - \chi_B|, 2\pi - |\chi_A - \chi_B|) \right)^2, \quad (2)$$

using the geodesic distance the offset operator already relies on for $\hat{n}$, and the circular distance a phase requires for χ ; r alone is a bounded line with no wraparound and needs neither correction. Raw coordinate subtraction on $\hat{n}$ or χ is *not* a safe default: two directions or phases near 0 and near 2π are adjacent on the sphere or the circle but appear maximally distant under naive subtraction, and reference atoms placed near a natural zero point — which is common, not rare — would be misjudged systematically rather than occasionally. *Open in this paper:* the weights $w_r, w_{\hat{n}}, w_{\chi}$, meaning how much a mismatch in strength, direction, or phase should count against an otherwise close match. §5 does not need this choice made in general: every shell and every reference point there already shares $\hat{n}_q$ by construction, so $w_{\hat{n}}$ never enters, and the verdict is read from χ (which reference point is nearest) with r supplying the null rather than a weighted distance. A question with reference atoms at different directions would need $w_{\hat{n}}$ and w_{χ} actually chosen against each other, which remains open.

A similarity metric. $d(A, B)$ answers how far apart two atoms are; the same question read the other way round — how alike — has never been named, though the raw ingredient has been sitting inside Equation (2) since it was written. Direction similarity is exactly the dot product the geodesic term already computes, before $\arccos$ turns it into an angle:

$$\text{sim}(A, B) = \hat{n}_A \cdot \hat{n}_B. \quad (3)$$

Bounded $[-1, 1]$ by construction — +1 identical direction, 0 orthogonal, −1 antipodal — and exactly invertible against the geodesic term already in use: $\arccos(\text{sim}(A, B))$ recovers that term precisely, confirmed to four decimal places across a range of directions. This is deliberately the narrow, direction-only reading, matching how cosine similarity is ordinarily used elsewhere (comparing embedding directions, not a magnitude-weighted blend); it needs none of $w_r, w_{\hat{n}}, w_{\chi}$, so it is available today regardless of how that open choice is eventually settled. A fuller similarity built from the whole of $d(A, B)$ — such as $e^{-d(A, B)}$ — would answer a related but different question, weighting strength and phase alongside direction, and would inherit the same open weights $d(A, B)$ itself depends on; left for whichever setting first needs that fuller reading rather than adopted here.

4.2 A selection rule

For Reading B, the order π that the evidence is applied in is handed to the walk; it is not the walk’s to choose. A real system must decide which signs enter B at all (§4.1’s window

and radius criteria, companion paper §2.2) and, when order matters, in what sequence. This selection is itself where most of an inference system’s actual difficulty lives, and it is entirely outside the algebra. *Open in this paper.*

A sequence is a degenerate graph. A walked path $T_{w_n} \circ \dots \circ T_{w_1}(x)$ is a simple directed path, one actor after another with no branching. When π is not given in advance, the natural structure is not a single path but a tree of candidate paths: several possible next actors branch from a shared prefix (twinkle, twinkle, then *either* little or something else), and choosing a walk means searching that tree rather than replaying one. Every edge of the tree is still one application of T ; the tree organizes a search over sequences, it does not add an operator. How candidates branch, how a branch is scored before it is fully walked, and how the tree is pruned are the actual content of this open selection rule, not resolved by naming the structure.

4.3 A principled null

Reading A supplies one for free: $z_B = 0$. Reading B does not. A walk always lands somewhere, and a comparison rule (§4.1) always returns a nearest pin, however far away it is. A Reading-B system therefore needs an explicit distance threshold beyond which the verdict is *abstain* rather than *nearest pin regardless*. This is a second place, distinct from §4.1, where a number must be chosen rather than derived.

4.4 Learned pins and learned signs

The companion paper notes that a lookup E_L may be learned (§5.6) without the units it points into being anything other than fixed. The same discipline applies to inference: what is learned is narrow. A sign’s (r, χ) for a given question — how strongly, and in which direction, a particular bird’s flight bears on “storm” — is an E_L entry, learnable from labelled cases. The seated pins storm, calm are likewise E_L entries, and may be learned rather than declared. Neither requires learning a representation in the sense of an embedding; both are learning one small map, per sign, per question. *This is the paper’s one concrete claim about the learning loop:* it is cheaper than representation learning because it is narrow, not because it is avoided.

Building a reference atom. A pin need not be declared; it can be built from witnessed cases the same way live evidence is read. For a class c (one storm, one drought), let $G_c = \{y_1, \dots, y_m\}$ be the Reading-A composite of each witnessed episode, seated by E_L from that episode’s own bowl and already rotated onto the question’s axis $\hat{n}_q$ (§3.2). Because every y_i shares $\hat{n}_q$ by construction, $\boxplus(G_c)$ is defined by the companion paper’s bowl-Mix equation (§2.7) without needing its still-open case of Mix across differing directions (its Future Work); building a reference atom this way never depends on a question that paper leaves unresolved. The reference atom for c is $\text{ref}_c = \boxplus(G_c)$, its strength reported by Equation (1) rather than the companion paper’s clip, per §2.1.

Coherence, and when to split. The capped strength $r(\text{ref}_c)$ is not a safe coherence signal on its own: it saturates at $\min(1, |z_{G_c}|)$, so a tightly clustered class and a widely scattered one can both report the maximal value once enough witnesses push $|z_{G_c}|$ past 1, at exactly the point where the underlying spread most needs distinguishing (§4.4 makes

this concrete). The uncapped magnitude alone does not fix this either, since it says nothing about how much evidence produced it: $|z_{G_c}| = 0.6$ means one thing from a single witness and another from nine that largely cancelled.

The correct coherence signal normalizes by both at once — the *mean resultant length*, the standard measure of angular concentration for a set of phasors:

$$\bar{R}_c = \frac{|z_{G_c}|}{\sum_{y \in G_c} r_y}, \quad z_{G_c} = \sum_{y \in G_c} r_y e^{i\chi_y}, \quad (4)$$

using the same uncapped z_{G_c} the companion paper’s bowl-Mix equation already computes before capping, alongside one further sum, $\sum r_y$, tracked once when G_c is assembled and not otherwise needed for reading live evidence. $\bar{R}_c \in [0, 1]$: 1 for perfectly aligned witnesses, 0 for complete cancellation or uniform scatter, and — unlike $r(\text{ref}_c)$ — unaffected by how many witnesses there are or how strong each one is, so a class of nine and a class of ninety are judged by the same scale. This is not a new invention; $1 - \bar{R}_c$ is the standard circular variance, and the same quantity underlies tests (Rayleigh’s, for one) of whether a set of angles is distinguishable from uniform scatter at all — a route to choosing θ by a stated significance level rather than by feel, not adopted here but available.

The direction is a pointer, not only the magnitude. $\bar{R}_c$ reads the magnitude of z_{G_c} ; the direction, $\chi(\text{ref}_c) = \arg(z_{G_c})$, is a separate claim, and it degrades gracefully rather than failing outright as witnesses scatter. Simulated against a fixed true direction with $N = 8$ witnesses and increasing angular noise, mean error grows smoothly — 2.8° at low scatter, 15.1° , 29.1° , 50.7° at severe scatter — never a cliff, and continues rising toward, not past, 90° , the expected angle to a genuinely random direction, as scatter grows without bound: the mean estimate never becomes actively anti-correlated with the truth, though any single noisy draw still can be. It also sharpens with more witnesses at fixed noise: $N = 2$ gives 36.6° mean error, $N = 128$ gives 5.1° , improving smoothly the way an ordinary law-of-large-numbers estimator should. None of this is new either — $\chi(\text{ref}_c)$ is the circular mean direction, the direction-side counterpart to $\bar{R}_c$ ’s magnitude, both read off the same z_{G_c} .

The two claims must not be conflated. A low $\bar{R}_c$ does not make $\chi(\text{ref}_c)$ worthless, only uncertain — a badly scattered group can still point roughly toward the truth, the way 50.7° of mean error is still far closer to the truth than to 90° of no information. But a single group’s plausible-looking direction is not, by itself, evidence enough to trust: the same noise that produces a small average error also produces individual draws that land unusually close to the truth by chance, indistinguishable, from one reading alone, from a genuinely coherent source. What separates the two is repetition — several independent groups’ directions agreeing with each other is exactly the claim $\bar{R}_c$ already tests, applied one level up, over a bowl of directions rather than a bowl of witnesses.

Given a coherence threshold $\theta \in (0, 1)$ and a minimum group size $k_{\min}$ below which a split is not attempted: if $\bar{R}_c \geq \theta$, keep $\text{ref}_c = \boxplus(G_c)$; otherwise, if $|G_c| \geq k_{\min}$, partition G_c into sub-groups and recurse on each, so that a single label may resolve into several reference atoms (a slow-onset and a flash drought, say); otherwise accept ref_c as weak, or fall back to comparing incoming evidence against G_c ’s individual members rather than their compression. How G_c is partitioned when a split is warranted is not specified here and is left open, as is the choice of θ itself; θ must be bounded away from 0, since every singleton group has $\bar{R}_c = 1$ trivially and a threshold with no floor never stops splitting.

Group size strictly decreases at each split, so the recursion terminates regardless of θ 's exact value.

This is level closure, applied to references. A class discovered to need splitting is a level-2 object in its own right (companion paper, §2.2): the split is discovered from G_c 's own internal agreement, not asserted in advance, and each resulting sub-reference is exactly as usable — comparable, walkable, further splittable — as the class it came from. A domain with genuinely few witnesses (two droughts, not nine) is not a failure of the procedure; $k_{\min}$ says plainly that there is not yet evidence to subdivide responsibly, and the honest output is a weaker reference or a fall back to individual comparison, not a manufactured split.

Repeated splitting is a tree, and classifying against it means descending it.

A label that splits once and then splits again within a child produces a tree of reference atoms rooted at the original class: storm at the root, perhaps slow onset and flash as children, each itself subject to the same $\bar{R} \geq \theta$ test on its own witnesses. Comparing an incoming atom against such a tree is naturally top-down: match against the root's siblings first (storm, drought, rain, flood), then, within whichever is nearest, descend to its children if it has any, and repeat. This is the shape of a classical decision tree, but not its mechanism: a decision tree tests a raw coordinate against a learned threshold at each node; a node here compares a full state against a reference atom by Equation (2), using operators and a metric the algebra already supplies rather than introducing threshold tests as a new primitive. The tree is discovered by $\bar{R}$, level by level, the same way §4.4 discovers it, not designed in advance; the comparison rule and the null of §4.1 and §4.3 apply at every node unchanged.

Merging groups. Two witnessed groups of the same τ , of any sizes, merge exactly, without revisiting either group's raw witnesses: z_{G_c} and $\sum r_y$ are sufficient statistics, in the standard sense — each additive over a disjoint union, so

$$z_{\text{merged}} = z_{G_1} + z_{G_2}, \quad \left(\sum r_y\right)_{\text{merged}} = \left(\sum r_y\right)_{G_1} + \left(\sum r_y\right)_{G_2}, \quad (5)$$

give $\bar{R}_{c,\text{merged}}$ and $\text{ref}_{\text{merged}}$ identical to recomputing Equation (4) from the two groups' combined raw membership — checked to full floating-point precision on groups of size 5 and 9. Group size needs no special handling: it is already carried inside $\sum r_y$, so a group of 400 correctly dominates a group of 3 in the merged result without any explicit weighting step. The cost of merging is therefore $O(1)$ in the number of witnesses, holding only the two summaries rather than either group's membership.

Two witnessed groups of *different* τ do not merge this way — averaging a storm phasor sum against a drought one asserts a relationship comparison and Mix both already refuse, since both are τ -gated. The companion paper's own Axiom 4 already supplies the right mechanism for this case: gather is sort-blind, so the two groups combine losslessly into one bowl regardless of sort, preserving both types distinctly rather than fusing them; any further reading — $\bar{R}_c$, ref_c , Equation (5) — is then applied separately within each τ present in the combined bowl, exactly as it already is for any mixed-sort bowl. Same-sort merging needs a new identity because $\bar{R}_c$ is new to this paper; cross-sort merging needed none, since $\oplus$ was never sort-restricted to begin with.

Illustration. Nine witnessed episodes at $r = 0.6$ each, at $\chi = 8^\circ, 9^\circ, 9^\circ, 10^\circ, 10^\circ, 10^\circ, 11^\circ, 11^\circ, 12^\circ$, give $\sum r_y = 5.4$ and $\text{ref} = (1.00, \hat{n}_q, 10.0^\circ)$: capped $r' = 1.00$ and $\bar{R} = 0.9998$ agree — coherent, confidently kept. The same nine witnesses spread instead across $\chi = -40^\circ, -20^\circ, -10^\circ, 0^\circ, 10^\circ, 20^\circ, 30^\circ, 45^\circ, 60^\circ$ give the identical capped $r' = 1.00$, since $|z_{G_c}|$ still exceeds the cap — but $\bar{R} = 0.8689$, correctly showing this class as less coherent than the first even though the capped reading cannot tell them apart. Five witnesses at $\chi = 0^\circ$ and four at $\chi = 180^\circ$ — two real sub-types wearing one label — give capped $r' = 0.60$ and $\bar{R} = 0.11$: both signal a problem, but $\bar{R}$ states it as a small fraction of the evidence actually agreeing, not merely a moderate reading. A perfectly even split, five and five, gives $\bar{R} = 0.00$, the same cancellation that gives §5 its abstain, now flagging the class itself rather than a single reading of it.

4.5 A walk-based classifier

Everything above classifies by comparing a state against a *compressed* reference built via $\boxplus$. The same question — what does new evidence resemble — has a second mechanism, using Reading B instead of Reading A, that has been available since §3.1 and simply never assembled for this purpose: walk new evidence to a landing, $x = \text{land}_\pi(q)$, and compare that landing against a stored reference by Equation (2) — exactly what the companion paper’s storm section already gestures at (“an endpoint may fall near the storm pin or the calm pin”), now made precise by the metric §4.1 supplies rather than left as an undefined “near.” Nothing here is new machinery; it reuses Reading B and the comparison rule exactly as they already stand.

Why this classifier does not compress. What it cannot do, for now, is combine several witnessed landings into one reference the way §4.4 combines several Reading-A composites into ref_c . Every y_i feeding a Reading-A reference atom shares $\hat{n}_q$ by construction — Reading A’s own rotation step guarantees it (§3.2). A T -walk’s landing carries no such guarantee: nine witnessed episodes, walked independently through their own sequences of actors, generally land nine different places. Compressing those landings via $\boxplus$ would need exactly the companion paper’s still-open case of Mix across differing directions (its Future Work), or a combination rule for rotations this algebra does not currently define — averaging rotations is a real, known operation (a Karcher mean on $\text{SO}(3)$, say) but importing one is a decision for later, not assumed here.

The fallback is the whole mechanism, not a special case. A walk-based classifier therefore keeps every witnessed landing individually and compares a new landing against each — the same fallback §4.4 already names for a class too small to split (“fall back to comparing incoming evidence against G_c ’s individual members”); here it is the only available mechanism rather than a fallback from a failed compression. A single declared pin, as in the storm example, is simply the case of one witness kept rather than several.

Illustration. Seat the question at the north pole, $\hat{n}_q = (0, 0, 1)$. Two actors walk it: $r_{w_1} = \frac{1}{3}$ ($\alpha = 60^\circ$) about the x -axis, then $r_{w_2} = \frac{1}{3}$ ($\alpha = 60^\circ$) about the y -axis, landing at $\hat{n} = (0.433, -0.866, 0.25)$, i.e. $(\theta, \phi) = (75.5^\circ, -63.4^\circ)$. Two declared pins: dry at $(\theta, \phi) = (80^\circ, 300^\circ)$ and rain at $(\theta, \phi) = (30^\circ, 90^\circ)$. The geodesic distance (Equation (2), $w_r = w_\chi = 0$) from the landing to dry is 5.6° ; to rain, 102.5° . The verdict is dry, by

a wide and unambiguous margin, using nothing beyond a walk already defined and a metric already stated.

Blocked for later, not solved here: a genuine compressed reference path, standing in for several witnessed walks the way ref_c stands in for several Reading-A composites, needs either the cross-direction Mix question resolved or an explicit rotation-averaging operator adopted and justified on its own terms. Both are left for iteration.

5 Worked Example: The Diviner’s Shells

We work Reading A in full, since the companion paper never carries a phasor reading past two states, and its property demonstration (non-associativity, on an anonymous triple) is not a reading of a bowl that means something.

Seat the question *will it rain* at $\hat{n}_q$. Each shell, once cast, is read by a diviner (or, later, by a learned E_L) as a state (r, χ) at $\hat{n}_q$: $\chi = 0$ favors rain, $\chi = \pi$ favors no rain, and intermediate χ is a hedge. Five casts, each shell carrying $r = 0.4$ unless noted:

Cast	reading	interpretation
4 shells, all favor rain	$r = 1.00$ at $\chi = 0^\circ$	confident rain
3 favor, 1 against	$r = 0.80$ at $\chi = 0^\circ$	rain, less confident
2 favor, 2 against	$r = 0.00$	abstain
4 favor, 2 hedge at 90°	$r = 1.00$ at $\chi \approx 26.6^\circ$	rain, hedged direction
12 shells ($r = 0.15$ each), all weakly favor	$r = 1.00$ at $\chi = 0^\circ$	confident rain

The third row is the case worth dwelling on. Two shells favor rain and two oppose it, at equal strength; the phasors cancel exactly, and the verdict is black — not a coin flip, not a default to the majority class, but the algebra’s own statement that this evidence does not decide the question. A system built on a forced classifier (softmax over two classes, say) has no way to express this outcome; it must output a confident-looking number even when the evidence is silent. Reading A’s null is not an afterthought fitted onto the algebra; it is Axiom 2, doing here exactly what it does for a Mix reading of colour.

The fifth row shows the saturation the companion paper’s §2.7 discusses: twelve weak, agreeing signs pin at $r = 1$ exactly as four strong ones do, and the difference between “moderately confident” and “overwhelmingly confident” is lost. This is the place where §4.2’s selection rule matters most: whether those twelve readings are twelve units at one level, or repeated samples of one unit that should have been reduced (companion paper, Axiom 4) before ever reaching the bowl, is a question this paper does not resolve and flags as the clearest candidate for the next.

§2.1’s substitution softens this specific symptom without resolving the question just raised. Calibrated to this domain’s evidence with $K = 0.2$, the fourth row reads $r' = 0.899$ and the fifth $r' = 0.900$ — genuinely distinguishable, where the clip read both as 1.00 — and even a hundred thousand such weak witnesses would read $r' = 0.99999$, not exactly 1. What is now settled is that the two *can* be told apart; what remains open is whether twelve weak signs are honestly twelve co-present witnesses of one thing, or repeated samples of a single witness that should have been reduced before reaching the bowl at all. The substitution reports strength faithfully either way and does not decide which reading was owed.

Reading B on the same shells is not worked here. Doing so honestly requires an order for the casts, which a physical handful of shells thrown at once does not supply — exactly the case the swap test (§3.3) assigns to Reading A. A worked Reading B example belongs to evidence that genuinely arrives in sequence, and the companion paper’s storm already provides one.

6 Level Closure in Practice: Correcting a Path

A T -path records the sequence of actors applied to it, faithfully, by definition (companion paper, Axiom 6). It has no means of knowing whether that sequence was the right one — level closure (§2) says precisely this: a lower unit carries no representation of what it constitutes, and neither the atom nor the path can see the higher-level context that would tell it it has erred.

A concrete case: a sequence of observed signs is walked in the order observed, but a later sign (or a completed higher-level unit, such as the rest of a sentence or the rest of a song) makes that order appear wrong — *twinkle, little, twinkle, star* rather than *twinkle, twinkle, little, star*. Correcting this is a level-2 operation: it requires comparing the recorded path against context the path itself does not have access to, and revising the walk probabilistically. This paper’s position is that the path must therefore be *legible* to the level above — readable as an ordered record, not only walkable — so that a corrector at the next level can inspect and revise it. Nothing in the companion paper’s axioms prevents this; nothing in them provides it either. We flag it as the clearest point of contact between this paper’s verdicts and a learning loop operating above them, and leave the corrector itself unspecified.

7 Prediction

Everything above compares an atom to a reference built from atoms of the *same* concurrent identity: a storm reference is built from patterns that were themselves storms, and a match says what the evidence currently resembles. That is classification. Prediction — what is likely to happen, not what this already is — needs a different labelling convention, not a different mechanism.

7.1 Immediate forecast: relabelling by outcome

Fix a lead time Δ . A *precursor witness* for outcome c is a Reading-A composite y_i of the evidence bowl observed at some time t , labelled not by what it is but by what followed it: c occurred at $t + \Delta$. Gathering such witnesses gives a precursor group G_c^Δ , and everything in §4.4 applies unchanged: $\text{ref}_c^\Delta = \boxplus(G_c^\Delta)$, checked for coherence by $\bar{R}_c$ (Equation (4)) and split if c conceals more than one precursor shape, exactly as §4.4 splits drought. Forecasting is then comparison against $\{\text{ref}_c^\Delta\}$ by the ordinary rule of Equation (2): the nearest precursor reference, within threshold, says what is likely at $t + \Delta$. The null of §4.3 applies exactly as before — no close precursor match is an honest *no forecast*, not a forced one, which matters most exactly where a wrong call is costly.

Nothing here is a new equation. The only change from §4.4 is which witnesses are gathered under a label: concurrent identity for classification, a fixed time-shifted outcome for this. Two things mark it as a first, crude pass rather than a finished one, named so

they are not mistaken for settled choices: Δ is fixed and single-valued, so the forecast has one horizon and no notion of a nearer or more distant event both being plausible; and no forecast carries a horizon-dependent confidence — a match found for $\Delta = 10$ minutes and one for $\Delta = 10$ days are read with the same $\bar{R}_c$ scale, though the latter should plausibly be trusted less for the same coherence value. Both are left for iteration, not resolved here.

7.2 A cheap decoder: transition counting over actors

Everything above answers “what does this resemble.” A different question — “what comes next in an unfolding walk” — has a mechanism simpler than either technique in §7.4, and it is worth stating precisely because it needs none of the machinery either one does: no metric, no $\boxplus$, no coherence check. It is a first-order Markov chain over the discrete vocabulary E_L already provides — the same technique, under the same name, used since the earliest work on statistical language modelling.

Given a set of witnessed trajectories, each already an ordered sequence of actors $(w_1, \dots, w_n)$ (companion paper’s pipeline equation, §2.2), count how often each actor is followed by each other across every witnessed path:

$$\hat{P}(w_{k+1} = w' \mid w_k = w) = \frac{\text{count}(w \rightarrow w' \text{ across all witnessed paths})}{\sum_{w''} \text{count}(w \rightarrow w'' \text{ across all witnessed paths})}. \quad (6)$$

Decoding is a lookup: given the actor just visited, predict the w' maximizing $\hat{P}$, or sample from it. Nothing here touches $T_w(x)$; the mechanism operates entirely on *which* actor follows *which*, never on the rotation either one performs. It is closer to bookkeeping on the tape a walk already produces than to a use of the walk’s own geometry.

Illustration. Four witnessed paths — (twinkle, twinkle, little, star) twice, (twinkle, little, star) once, and (twinkle, twinkle, little, bat) once — give

$$\hat{P}(\cdot \mid \text{twinkle}) = \left\{ \text{twinkle} : \frac{3}{7}, \text{ little} : \frac{4}{7} \right\}, \quad \hat{P}(\cdot \mid \text{little}) = \left\{ \text{star} : \frac{3}{4}, \text{ bat} : \frac{1}{4} \right\}.$$

A single witnessed path alone would have given an uninformative $\frac{1}{1}$ split; the distinction above only exists because several paths were witnessed, not one.

What this mechanism does not use, stated plainly. The atom carries $r, \hat{n}, \chi$; this decoder reads none of them, only τ -gated discrete identity. For this specific mechanism, a bare token would do exactly as well — the geometry is present but idle. It is named here rather than left implicit, since claiming otherwise for this particular technique would overstate what it uses.

Cost, stated honestly. It needs several witnessed repetitions before its output means anything, and it only ever looks back a short, fixed window (one prior actor here; an n -th order chain extends the window at the cost of needing correspondingly more witnessed data to fill it) — it has no notion of the deeper structure a full prefix match, §7.3 below, would catch. Between the two, this is the cheaper, unblocked mechanism for the same job; the exemplar-based one is the deeper, still-open alternative.

7.3 Open: path predictors

The immediate forecast reads only the endpoint of what happened after a precursor; a deeper forecast reads the whole unfolding. Store, per witnessed episode, the entire trajectory that followed — not one composite but the ordered sequence of atoms, timestamped as the companion paper’s step-metadata generalization allows (§3.3) — and match an *observed prefix* of a trajectory now against the opening steps of stored trajectories, reading the continuation of the closest match as the prediction. This is not a new idea invented for this paper: it is the analog method already used in operational forecasting, arrived at here from the algebra’s own machinery rather than borrowed from it.

It is filed open, not attempted, because it needs two things this paper has not built. First, storage of complete historical trajectories rather than single compressed reference atoms, a real cost the immediate forecast does not pay. Second, and more substantial: Equation (2) compares one state to one state, and prefix matching compares a *sequence* of states to a sequence, which needs a distance over sequences — some alignment or step-wise accumulation of the single-state metric, robust to two episodes unfolding at different paces — that has not been defined. This section marks where that work belongs rather than attempting it; the immediate forecast of §7.1 does not depend on it and should not wait for it.

7.4 Four cells, two techniques

Comparison (§4.1) is the one operation behind everything in this section and the last — everything, that is, except §7.2’s transition count, which answers a different question (what comes next, not what this resembles) without comparison at all. Two established techniques are built from comparison, depending on what a produced state is checked against: *nearest-centroid classification*, comparing against a single reference compressed from many witnesses via $\boxplus$ (§4.4), and *nearest-neighbor classification*, comparing against witnesses kept individually (§4.5). Neither is specific to this paper — both are standard, decades-old techniques in pattern recognition, arrived at here from the algebra’s own constraints rather than imported by name.

A second, independent axis is what the label means: identity, giving a verdict about now, or a time-shifted outcome, giving a forecast about later (§7.1). The two axes cross to give four cells, not a new technique for each:

	now (identity label)	later (outcome label)
nearest-centroid	§4.4	§7.1
nearest-neighbor	§4.5	§7.3

The bottom-right cell already has its own established name for the same reason the top row does: nearest-neighbor comparison, run against outcome labels, over whole sequences rather than single states, is the analog method §7.3 cites.

The split between rows is not stylistic; it is forced by the same fault line as $\boxplus$ and T . Compression into a centroid is available wherever $\boxplus$ ’s composites are guaranteed to share a direction; keeping individual neighbors is what remains wherever that guarantee fails — a T -landing (§4.5), or, in the one open cell, a whole sequence with no defined distance yet.

Three of the four cells are specified and checked on worked numbers; none should be read as more finished than that. Nearest-centroid classification (§4.4) is complete as a

mechanism, with three choices left named and open within it: the coherence threshold θ , the comparison weights $w_r, w_{\hat{n}}, w_\chi$, and how a class is partitioned once a split is warranted. Nearest-neighbor classification (§4.5) is complete for what it does — walking new evidence and comparing the landing against kept witnesses — with compression into a single centroid blocked separately, not left unfinished. The immediate forecast (§7.1) is explicitly a first pass: one fixed lead time Δ , no horizon-dependent confidence, both named for iteration rather than resolved. The fourth cell, §7.3, is open outright, waiting on the sequence-level distance neither this table nor that section supplies. None of the three built cells has been evaluated against real data; each is verified on the toy numbers of its own illustration, per the framework-versus-benchmarked choice §10 leaves open.

8 Assembly: A Second Structural Verb (Status Report)

Read this paragraph before anything else in this section. Everything below was worked out and numerically verified in conversation, not through the same drafting and review process as every section preceding it. None of it has been held to the citation discipline, the cross-checking, or the settling time the rest of this document received before being written down. It is recorded here specifically so its state is legible — what is closed, what is open, and what is only a named gap with no attempt yet — not because any of it is ready to stand as a finished part of Verdicts. Nothing in this section should be cited or relied upon as if it carried the same weight as §4.1 through §7.2.

8.1 What is settled: the two-body case

A second technique, distinct from comparison, was explored: two atoms combining not by fusion ($\boxplus$) and not by bare co-presence ($\oplus$), but by a relation added while both remain intact.

- **Signature.** A Assembly $B = (A, B, \kappa_{AB}, U_{AB})$ — both atoms preserved, a contact point and an interaction energy added.
- **Energy, closed form.** $U_{AB} = -\frac{r_{AB}}{2}(3 + \hat{n}_A \cdot \hat{n}_B)$, a single dot product, unconditionally defined at every angle including exactly 180° . Verified against an axis-based derivation to floating-point precision (10^{-15}) over 20,000 random pairs.
- **Contact point.** $\kappa_{AB} = \hat{n}_A + \hat{n}_B$, the raw, unnormalized sum — always defined, its own magnitude a free confidence signal.
- **The antipodal case.** Resolved, not patched: the energy formula above never needed an axis at all, so the degeneracy that motivated days of exploration (a bi-sector dividing by zero) turned out to be an artifact of an unnecessary normalization step, not a property of the underlying relationship.
- **Repulsion is impossible for an isolated pair.** Proven, not assumed: checked against 100,000 random pairs of varying orientation and strength; the least attractive result found was still negative.
- **Group energy.** $\sum_{\text{pairs}} U_{ij}$, the standard physical form for total multi-body interaction energy — computable today for any bowl.

- **Group reconciliation, closed.** The original worry — an atom cannot simultaneously occupy several private, independently-optimal axes with different neighbors — dissolves once real position exists. With position, no axis is being optimized per pair at all; each pairwise force is simply read off real geometry, and total force is their ordinary vector sum, $F_{\text{total}} = \sum_i F_i$. Verified directly on a three-atom case: no ambiguity, no conflict, nothing left to reconcile beyond arithmetic already available.

8.2 Position and distance

Each atom may carry an ordinary position vector $\vec{P}_A$, measured from one shared origin — deliberately external to the bounded $r \leq 1$ stage, the same way time already sits outside the atom on the path rather than inside it.

- **Distance.** $d = |\vec{P}_B - \vec{P}_A|$, plain Euclidean distance.
- **The real axis supersedes the derived one.** Once real positions exist, the separation axis is $(\vec{P}_B - \vec{P}_A)/d$, computed from position, not derived from orientation. Checked directly: two atoms left exactly antipodal in orientation, given distinct positions, produce a perfectly well-defined axis and energy — the antipodal problem was never triggered in the first place once position exists. The earlier orientation-derived axis (§8.1) remains useful only as a fallback when position is unknown.
- **Falloff.** The standard $1/d^3$ scaling was used throughout; not derived here, borrowed directly from the physical dipole formula this whole construction has tracked.
- **Dynamics, closed.** Motion follows Axiom V4: gradient descent, no volume term, no momentum. Verified stable at $N = 2, 3$, and 8, and confirmed as the deliberate, final rule rather than one candidate among several — volume-weighted motion was tested as the alternative and shown to destabilize precisely where volume ratios grow extreme, which settled the choice on stability grounds rather than convenience.
- **Symmetric or actor/patient, closed at criterion, complete by design.** Polarity (§8.4) supplies the criterion — equal strength suggests a mutual bond, unequal strength suggests an actor and a patient, mirroring T 's own asymmetry. No further behavioral mechanism is missing: it was tested directly and found to require exactly the volume-weighted motion Axiom V4 excludes on stability grounds. The honest, final scope of this question is a label describing a bond, not a difference in how it moves.
- **Identical-atom multiset, closed.** A assembling with an exact copy of itself needed a count, not a set — Axiom V5 supplies it: $\{A, A\}$ collapses under ordinary set equality, $m(A) = 2$ does not. Two genuinely separate witnesses sharing an identical tuple are now distinguished by multiplicity rather than lost to it, without any change to Axiom 4 or $\oplus$ itself.

8.3 What remains open

- **Mass segregation (the astrophysical phenomenon), superseded as a prediction here.** Named early on as an expected consequence while this document's own dynamics was still built around a mass-like quantity: stronger members sinking

toward a group’s center, weaker members toward the periphery, as in real globular-cluster relaxation. Once Axiom V4 settled motion as excluding $\mathcal{V}_{\text{atom}}$ entirely, this prediction’s own mechanism no longer applies here — confirmed directly, equal force with nothing dividing it gives identical movement regardless of strength or volume. The astrophysical phenomenon keeps its real name; nothing in this framework currently reproduces it, since the quantity that would need to drive it was deliberately kept out of motion. Recorded as a real idea that depended on a choice later reversed, not as something still to be verified.

- **Multi-vector generalization, largely closed.** Every atom keeps its N informational vectors, each capped at $r \leq 1$ as usual and counted toward atomic number; alongside them, in the same closure unit, sits one derived vector — $\sum$ of the informational vectors, uncapped, excluded from the count, needing no τ -justified admissibility of its own, the same status as $\bar{R}_c$ or z_{G_c} . Coarse-grained assembly (the common case) runs on the derived vector, so U_{AB} and κ_{AB} never actually needed to generalize at all. What remains optional: fine-grained, simultaneous multi-partner bonding via the informational vectors directly, with two sub-mechanisms identified — an unconstrained double sum over every vector pair, or an exclusive matching (the assignment problem, solved via the Hungarian algorithm). Checked precisely where these agree: sorting each atom’s vectors by strength and pairing by rank matches the true optimal matching whenever alignment quality is roughly uniform across candidates (a consequence of the rearrangement inequality) but can be strictly worse than optimal when strength and alignment pull in different directions for different pairs — verified with a constructed adversarial case (-1.17 for rank order versus -1.68 true optimum). The honestly correct general statement: the lowest-energy pairing is the stablest, and other pairings are unstable and will dynamically move toward it; strongest-with-strongest is that stable destination only in the uniform-alignment special case, not in general.

8.4 Derived features

Several quantities were derived from the existing atom, aggregate, or pair, following the discipline already established for $\bar{R}_c$ (Equation (4)): computed, not assigned, and checked against what gap, if any, they close.

Feature	What it is	Status
Atomic number	Vector count per atom, proposed as bonding capacity	Speculative; not adopted
Density	Strength divided by vector count	Speculative; depends on atomic
Energy ($\sum r_i$)	Already established (§4.4, §2.1)	Settled, predates this section
z_{G_c} , the vector sum	Raw, uncapped $\sum r_y e^{i\chi_y}$; direction plus absolute magnitude, neither carried by $\sum r_i$ or $\bar{R}_c$ alone	Verified; closes two real gaps (c
Polarity ($ r_A - r_B $)	Whether a pairwise bond is balanced or one-sided	Verified concept; supplies the c
Heterogeneity	Variance of strengths within a group, distinct from $\bar{R}_c$'s directional coherence	Named, not built at the bowl l
Volume concentration (σ_{atom})	How evenly an atom's volume is spread across its own informational vectors (Equation (9)); 0 even, 1 fully concentrated, independent of N	Verified; adopted in Axiom V1
Information energy ($\mathcal{E}_{\text{atom}}$)	Combined squared magnitude of an atom's own informational vectors as if mutually orthogonal (Equation (7)); defined directly, not as strength squared — the same relation variance has to standard deviation. Distinct from bowl energy, which sums across witnesses rather than within one atom	Verified; adopted in Axiom V1
Shell concentration (σ_{shell})	Whether many vectors independently cluster near the same magnitude (Equation (10)); a blind spot of σ_{atom} , confirmed directly — a clumped population read as more even than a uniform spread until this was built. Needs a stated shell width w	Verified; adopted in Axiom V1
Internal alignment (α_{atom})	Strength-weighted mean cosine similarity across an atom's own informational vectors (Equation (8)); signed, so it separates active cancellation from mere non-reinforcement	Verified; adopted in Axiom V1
True angular dispersion	Distinguishes genuine scatter from a tight two-cluster split, which $\bar{R}_c$ and z_{G_c} both read as zero either way	Named, not built; confirmed as
Path turning-angle variance	How smooth or erratic a walked sequence is	Named, not built

Checked cross-connections: polarity closes the criterion half of the symmetric/actor-patient question above; z_{G_c} 's mechanism and true angular dispersion are plausible, untested

leads toward group reconciliation; density has a conditional connection to group reconciliation if read as implying physical size. Atomic number does not close a gap — adopting it creates one (multi-vector generalization). Energy, heterogeneity, and path turning-angle variance were checked and found to close none of the open assembly items.

9 Higher Enclosure: Axioms for Position, Motion, and Multi-Vector Atoms

Status note. The four axioms below organize §8’s findings into the same axiomatic form the companion paper uses, so the accumulated work has a durable, citable shape rather than remaining a scattered set of verified claims. They are Verdicts-native — numbered V1–V4, deliberately not continuing the companion paper’s own Axiom 0–5, and nothing here is proposed for that document. Each has been checked numerically, several extensively, within this single sustained session; none has had the multi-session settling time every axiom in the companion paper received before being written down. Two items named in §8.3 — the identical-atom multiset, and the full behavioral realization of the actor/patient question — remain genuinely open and are not resolved by anything below.

9.1 Axiom V1: Informational and derived vectors

An atom’s informational vectors are N ordinary units of Axiom 1 (companion paper), each independently capped at $r \leq 1$; N is the atom’s info number. Alongside them, in the same closure unit, at most one *derived* vector may be present:

$$V_{\text{derived}} = \sum_{i=1}^N v_i,$$

the raw, unnormalized sum of the informational vectors, computed after and from them, never including itself. V_{derived} is uncapped and is excluded from N : it requires no τ -justified admissibility of its own, the same status already held by $\bar{R}_c$ and z_{G_c} (companion paper, §4.4, §7.4). Its magnitude and direction are read as an effective $(r, \hat{n})$ pair — $r = |V_{\text{derived}}|$, $\hat{n} = V_{\text{derived}}/r$ — standing in for the atom wherever a single vector is required. For $N = 1$, V_{derived} coincides with the atom’s own vector and nothing changes.

Several scalar metrics follow directly from the same N informational vectors, each answering a different question and none used by any other axiom below unless explicitly named:

$$S_{\text{atom}} = \sum_{i=1}^N r_i, \quad \mathcal{V}_{\text{atom}} = \sum_{i=1}^N r_i^3, \quad \gamma_{\text{atom}} = \frac{\mathcal{V}_{\text{atom}}}{N}.$$

S_{atom} , atom strength, is the correct denominator for atom-level coherence, $|V_{\text{derived}}|/S_{\text{atom}}$, the structural twin of $\bar{R}_c$ one level down — verified to read 1.0 for perfectly aligned informational vectors and fall smoothly toward 0 as they scatter. $\mathcal{V}_{\text{atom}}$, atom volume, is strictly smaller than S_{atom} whenever any $r_i < 1$, a forced consequence of the $r \leq 1$ cap rather than an assumption: cubing a number below 1 always shrinks it, exactly as treating r as a literal radius and r^3 as the volume it encloses would predict. γ_{atom} , mean vector volume, is the average per-vector share of that total. None of the three is used in Axiom V4; atom volume, in particular, is deliberately excluded from motion (Axiom V4) and exists here purely as a descriptive metric of the atom. Earlier drafts of this axiom

called $\mathcal{V}_{\text{atom}}$ “mass,” γ_{atom} “density,” and S_{atom} “atom energy” — all three retired as misleading: the first two because neither participates in momentum, kinetic energy, or any calculation the word “mass” would ordinarily license, and the third because a genuine notion of energy exists here and is not S_{atom} .

A fourth quantity is not derived from strength by squaring it, but stands on its own, the way variance is defined directly rather than as squared standard deviation. Writing each informational vector as $v_i = r_i \hat{n}_i$:

$$\mathcal{E}_{\text{atom}} = \sum_{i=1}^N r_i^2. \quad (7)$$

$\mathcal{E}_{\text{atom}}$ is the combined squared magnitude the atom’s informational vectors would carry if every pair were mutually orthogonal — confirmed by direct construction, computing $\mathcal{E}_{\text{atom}}$ from a set of strengths matches $|V_{\text{derived}}|^2$ computed from those same strengths placed at mutually orthogonal directions, exactly. This is forced, not chosen: an exact identity places $\mathcal{E}_{\text{atom}}$ inside the derived vector’s own squared magnitude,

$$|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j),$$

verified to six decimal places. The structure matches the variance of a sum, $\text{Var}(\sum X_i) = \sum \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$, term for term: r_i^2 stands where $\text{Var}(X_i)$ stands, each vector’s own self-contained contribution; $r_i r_j (\hat{n}_i \cdot \hat{n}_j)$ stands where $\text{Cov}(X_i, X_j)$ stands, how two vectors relate. $\mathcal{E}_{\text{atom}}$ is the part that remains when every cross term vanishes — the atom’s intrinsic energy, prior to any question of how its own vectors align. This is distinct from bowl energy ($\sum r_i$ across many separate witnesses, §4.4): bowl energy is a sum across atoms, $\mathcal{E}_{\text{atom}}$ a sum of squares within one.

A fifth metric answers a question the other four cannot. Atom coherence above is a ratio of magnitudes and so cannot fall below zero: it reads 0 both for informational vectors that merely fail to reinforce and for vectors that actively cancel. The strength-weighted mean cosine similarity across all internal pairs separates these:

$$\alpha_{\text{atom}} = \frac{\sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)}{\sum_{i < j} r_i r_j}. \quad (8)$$

Each $\hat{n}_i \cdot \hat{n}_j$ is already a cosine similarity, since informational vectors are unit-directed; the denominator makes the aggregate comparable across atoms of differing N and differing strengths, which the unnormalized sum is not — two atoms with near-identical internal arrangement but $N = 2$ and $N = 5$ give raw sums of +0.72 and +0.90 against normalized values of +0.995 and +0.998. $\alpha_{\text{atom}} = +1$ exactly when all informational vectors align; it is negative when they genuinely oppose, reading $-1/3$ for two opposed pairs. Its floor is not -1 in general: three or more vectors in three dimensions cannot be mutually opposed, so achievable disagreement tightens as N grows while achievable agreement does not — the unnormalized sum has maximum $\sum_{i < j} r_i r_j$, growing as N^2 , against a hard minimum of $-\frac{1}{2} \mathcal{E}_{\text{atom}}$, growing only as N , which follows directly from $|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j) \geq 0$. Agreement compounds; disagreement saturates. α_{atom} is read alongside atom coherence, not in place of it: the first asks whether the vectors agree, the second how much of their strength survives into one resultant.

A sixth metric does for magnitude what α_{atom} does for direction. S_{atom} and $\mathcal{V}_{\text{atom}}$ both report a total and are silent on how that total is spread: two atoms can carry near-identical strength (2.200 against 2.150 in a checked case) while one holds it evenly across

its vectors and the other concentrates most of it in two. Writing $s_i = r_i^3/\mathcal{V}_{\text{atom}}$ for the share of volume held by vector i :

$$\sigma_{\text{atom}} = \frac{N^2}{N-1} \cdot \text{Var}(s_1, \dots, s_N), \quad N \geq 2, \quad (9)$$

with $\sigma_{\text{atom}} = 0$ by convention at $N = 1$. $\sigma_{\text{atom}} = 0$ exactly when volume is spread evenly and 1 when a single vector holds all of it, at every N — verified at $N = 2, 3, 4, 6, 8$ for both extremes. The normalization is what earns this form over the simpler candidates: the largest vector's share and the Herfindahl index were both tested and rejected, since each reads a perfectly even atom as $1/N$ and so slides from 0.5 to 0.125 as N grows from 2 to 8, confounding evenness with size. $\mathcal{V}_{\text{atom}}$ rather than S_{atom} supplies the shares deliberately: cubing amplifies concentration, so a single dominant vector at $r = 0.95$ against three at $r = 0.2$ registers a share of 0.973, where linear strength would report the same atom as unremarkable.

A seventh metric answers a question none of the first six can: not how strength or volume is shared among distinct vectors, but whether many vectors independently arrive at nearly the same magnitude. σ_{atom} is blind to this by construction — four hundred vectors at an identical $r = 0.65$ each receive their own tiny individual share, indistinguishable to Equation (9) from four hundred vectors scattered across the full range; checked directly, a genuinely clumped population registered as *more* even ($\sigma_{\text{atom}} = 0.00032$) than a uniform spread of the same size (0.00214). Fixing this needs a shell width w , a named, configurable parameter rather than a derived one, defaulting to $w = 0.1$ — ten shells spanning $[0, 1]$. Binning the N informational vectors by $\lfloor r_i/w \rfloor$ into K occupied shells with populations $n_1, \dots, n_K$ and $p_k = n_k/N$:

$$\sigma_{\text{shell}}(w) = \begin{cases} 1, & K = 1 \\ \frac{K^2}{K-1} \cdot \text{Var}(p_1, \dots, p_K), & K \geq 2. \end{cases} \quad (10)$$

The $K = 1$ case is stated separately because the general form has a removable singularity there: every vector sharing one shell is maximal concentration, not the zero the formula would otherwise return. $\sigma_{\text{shell}} = 0$ when occupied shells are evenly populated, at every K tested; $\sigma_{\text{shell}} = 1$ when every vector shares one shell. Checked at $w = 0.1$ and $w = 0.05$: well separated shells (as in 10@0.98, 3@0.3, 400@0.65, giving $\sigma_{\text{shell}} = 0.908$ at either width) are stable to the choice, while genuinely uniform populations are not — occupied shell count nearly doubled between the two widths in a checked uniform case, though the concentration score itself stayed negligible either way. w should be reported alongside σ_{shell} whenever it is used.

9.2 Axiom V2: External position

An atom, or a bowl, may carry a position vector $\vec{P}$, measured from one shared origin, in ordinary unbounded Euclidean space — deliberately external to the $r \leq 1$ stage, as time already sits external to the atom on the path rather than inside it. For two atoms A, B with positions $\vec{P}_A, \vec{P}_B$:

$$d_{AB} = |\vec{P}_B - \vec{P}_A|, \quad \hat{r}_{AB} = \frac{\vec{P}_B - \vec{P}_A}{d_{AB}}.$$

Where position exists, $\hat{r}_{AB}$ supersedes any orientation-derived separation axis; the latter remains a fallback only where position is unknown.

9.3 Axiom V3: Assembly

Two atoms combine by a relation added while both remain intact — distinct from $\oplus$ (bare co-presence, no relation) and $\boxplus$ (fusion, identities lost):

$$A \text{ Assembly } B = (A, B, \kappa_{AB}, U_{AB}),$$

$$U_{AB} = -\frac{r_A r_B}{2} \cdot \frac{3 + \hat{n}_A \cdot \hat{n}_B}{(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}}, \quad \kappa_{AB} = \hat{n}_A + \hat{n}_B, \quad \varepsilon_{AB} = r_A + r_B.$$

U_{AB} is unconditionally defined at every angle, including exactly antipodal orientation, and at every separation, including $d_{AB} = 0$. For an isolated pair, $U_{AB} < 0$ always: repulsion does not occur between two atoms considered alone (verified over 100,000 random pairs). For a group, total energy is $\sum_{\text{pairs}} U_{ij}$; reconciling several simultaneous relationships on one atom requires no separate mechanism once Axiom V2 holds, since each pairwise force is read from real position rather than independently optimized. r_A, r_B in the above are each atom's $(r, \hat{n})$ under Axiom V1 — an atom's own vector if $N = 1$, its derived vector otherwise.

9.4 Axiom V4: Motion without atom volume

Position changes by moving directly along the energy gradient. No term for $\mathcal{V}_{\text{atom}}$ appears, by deliberate design rather than by an unresolved choice: atom volume (Axiom V1) is a real, useful metric of an atom's own structure, but makes no sense as an input to motion, and is excluded from it.

$$\vec{P}_i \leftarrow \vec{P}_i + dt \cdot F_i, \quad F_i = -\nabla_{\vec{P}_i} \sum_{j \neq i} U_{ij}.$$

No inertia means no overshoot: an isolated pair closes distance monotonically and never exceeds its own starting separation (verified). A group of mutually attracting atoms converges to a stable configuration centered on $\vec{P}_{\text{center}} = \sum_i r_i \vec{P}_i / \sum_i r_i$, held stationary once reached rather than merely passed through — verified at $N = 2$, $N = 3$, and $N = 8$, flat to four decimal places over thousands of subsequent steps in each case. ε_{AB} (Axiom V3) must be sized relative to dt or the same instability a literal $1/d^3$ term would produce at close approach can recur.

A direct, checked consequence: since $F_A = -F_B$ exactly for any pair, and nothing divides the resulting step, two atoms of any polarity move by identical magnitude toward each other, regardless of strength. Weighting each atom's step by its own $\mathcal{V}_{\text{atom}}$ was tested directly as an alternative and produces the intended differential movement, but destabilizes into oscillation precisely where that ratio is most extreme — confirmed at a $9\times$ strength difference, giving a $729\times$ volume ratio and a demonstrably non-monotonic trajectory, against the clean monotonic decline of this axiom's rule under the same pair. Axiom V4 is therefore stated without $\mathcal{V}_{\text{atom}}$ on stability grounds, not merely convention; a differential-motion mechanism is not a matter this axiom leaves open, since any such mechanism reintroduces the instability directly.

9.5 Axiom V5: Multiplicity

A bowl's membership under the companion paper's Axiom 4 is binary: an atom is present or absent, since $\oplus$ is defined as ordinary set union on a subset of $\mathcal{S}$. Where the number

of times a value was independently witnessed must itself survive — not merely whether it occurred — a bowl may additionally be read through a multiplicity function:

$$m : \mathcal{S} \rightarrow \mathbb{N}_0, \quad m(A) = \text{the number of independent witnesses producing exactly } A.$$

An ordinary bowl is the special case $m(A) \in \{0, 1\}$ for every A , recovering the companion paper’s Axiom 4 exactly; nothing about $\oplus$ or Axiom 4 is altered, and m is read alongside a bowl, never in place of it. Where $m(A) \geq 2$, A is counted that many times in every aggregate drawn from the bowl — $\sum r_i$, $\bar{R}_c$ (Equation (4)), and any use of Assembly (Axiom V3) that draws on the bowl’s members individually. Two atoms sharing an identical $(r, \hat{n}, \chi, \tau)$ tuple, arising from genuinely separate witnesses, are distinguished by their multiplicity alone, never by any difference in the tuple itself: A assembling with an exact copy of itself is simply $m(A) = 2$, unremarkable once presence and count are kept as separate questions. Treating such a pair as the set $\{A, A\}$ is not a looser way of saying the same thing; $\{A, A\} = \{A\}$ under ordinary set equality, and a second, genuinely distinct witness is silently lost — confirmed directly: an atom of strength $r = 0.7$, witnessed twice, contributes 0.7 to $\sum r_i$ if read as a set and the correct 1.4 if read through m .

10 Scope and Open Questions

This paper is a framework, in the same sense the companion paper is: it specifies mechanism and states honestly what it has not demonstrated at scale. *[OPEN — decide before circulating: does this paper stop here, as a specification plus one worked toy example, or does it commit to an evaluation against real data (e.g. a labelled forecasting or forex task) with a stated baseline? The two choices imply different remaining sections and different claims in the abstract; see the companion paper’s own title history for how much that choice shapes what a title may honestly claim.]*

Regardless of that choice, the following are stated open rather than resolved by this paper:

- The comparison rule (§4.1) is a choice, not a derivation; a different metric may be more honest for a given domain.
- The selection rule (§4.2), including how the grouping window of companion-paper §2.2 is set, is the largest source of real difficulty and is entirely unaddressed here.
- The saturation–under–aggregation problem (§5, companion paper §2.7 and §2.3) is diagnosed, not solved: whether to route repeated samples through a reduction before the bowl, or revisit the companion paper’s Axiom 1 note on an order-preserving bijective bound, is left to whichever setting first needs to aggregate at scale.
- The path corrector (§6) is named and not built.
- Path predictors (§7.3) are named and not built; the immediate forecast (§7.1) does not require them.
- The coherence threshold θ and the method for partitioning a class once a split is warranted (§4.4) are open choices not fixed here; §7.4 consolidates this and every other item above by which of the four classification-and-forecast cells it belongs to.